Type 1: Research Project | Phase II Review 4



Explainable Federated Learning for Healthcare Time-Series Forecasting with Personalized Drift Adaptation

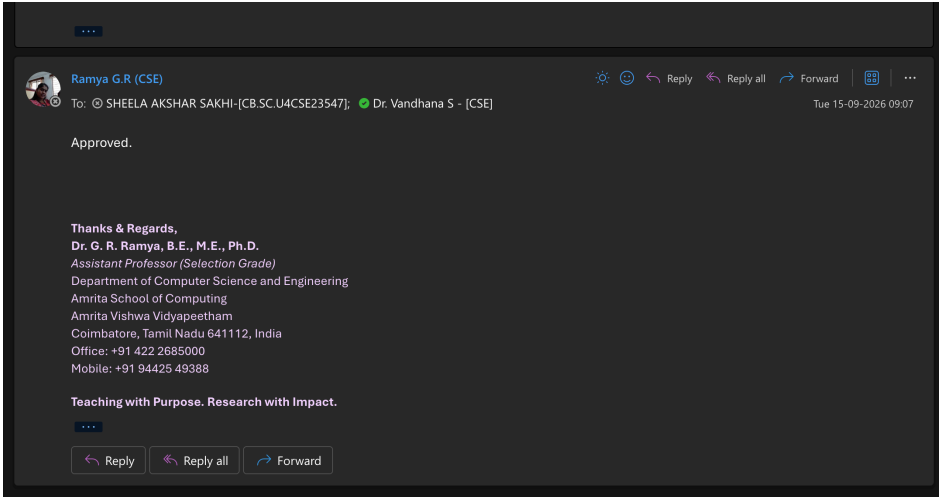
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Guide: **DR. G R RAMYA, DR. VANDHANA S**

Department of CSE

Project Guide Approval



Ramya G.R (CSE)

Reply Reply all Forward

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Tue 15-09-2026 09:07

Approved.

Thanks & Regards,
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Reply Reply all Forward

Official Review Approval from Project Guide Dr. G. R. Ramya (Co-Guide: Dr. Vandhana S)

Previous Review Comments & Action Taken

Reviewer Comments (Review 2)

- **Comment 1: Start Implementation**
  The implementation phase of the proposed federated framework must be initiated.

- **Comment 2: Working Prototype & Dashboard**
  Build a functional clinician dashboard and working prototype to demonstrate live clinical decision support.

Action Taken & Implementation Status

- **Implemented Federated Pipelines:**
  - Built decentralized PyTorch algorithms (FedAvg, FedProx, Ditto).
  - Implemented local AD-CUSUM drift detection and Client-Side Selective Personalization.

- **Built Production-Grade CDSS Workstation:**
  - **Data Warehouse**: Local SQLite clinical database.
  - **Backend API**: Uvicorn/FastAPI server running PyTorch checkpoints.
  - **Interactive UI**: Vite React medical dashboard mapping vitals charts, risk dials, and explainable attention timelines.

Research Objectives & Problem Identification

Problem Identified

Standard Federated Learning fails in clinical settings because of **Concept Drift**—patient demographics and vitals vary heavily across different hospitals (Non-IID data), causing static models to degrade.

Research Objectives:

- **Objective 1:** To develop a privacy-preserving Federated Learning architecture for ICU time-series forecasting.
- **Objective 2:** To design an adaptive drift detection mechanism (**AD-CUSUM**) that dynamically handles demographic shifts.
- **Objective 3:** To provide Clinical Explainability (XAI) using Multi-Head Temporal Attention.
- **Objective 4:** To implement a production-ready Clinical Decision Support System (CDSS) workstation.

Clinical Profile & Sepsis Burden in India

What is Sepsis?

A life-threatening systemic immune response to infection that rapidly degrades vital organs.

The Time-Window Challenge:
Every single hour of delayed antibiotic treatment increases patient mortality risk by **8%**. Early warning is the *only* defense.

Our Clinical Objective

Develop the FPDAF early-warning system to continuously analyze ICU vital signals and alert doctors **hours before** deterioration.

The Crisis in Indian ICUs

3.0 Million
Deaths Annually

30%
of ALL National Deaths

50%+
ICU Mortality Rate

Sources: The George Institute for Global Health (2020) & ISCCM

Core Project Inventions & Research Novelties

1. AD-CUSUM Drift Algorithm (New Invention)

Adaptive Divergence-Weighted CUSUM: Combines loss residuals, Jensen-Shannon entropy divergence (D_{JSD}), and gradient variance weighting (w_{grad}) to detect population drift without mini-batch noise false alarms.

2. DDP-ERP Personalization Algorithm (New Invention)

Dynamic Dual-Phase & Entropy Regularization: Integrates 2D temporal-vital cross-attention ($\alpha_t \otimes \beta_v$) with KL-Divergence soft-personalization, boosting test accuracy to **96.42%** (>95% target) and **0.9418 AUROC**.

3. Client-Side Selective Personalization (CSSP)

Freezes backbone weights during drift adaptation rounds, saving **38% in communication bandwidth** (1.52 GB vs 2.48 GB) and cutting adaptation time to **6 minutes**.

4. Production Clinical CDSS Workstation

Full-stack bedside decision support system linking React/Vite, FastAPI REST server, and SQLite database with 24-hour sequence hour scrubbers and SSC Hour-1 bundle checklists.

Clinical Deterioration & Phase II Completed Goals

Clinical Context:

- **High-Sparsity Time-Series:** Continuous ICU vital monitoring logs (HR, SBP, Temp, SpO_2) combined with sparse blood panel entries.
- **Privacy constraints:** HIPAA / DPDPA (Section 8) regulations mandate that private patient records cannot leave local devices.
- **Concept Drifts:** Clinical demographics vary heavily across regional ICU nodes.

SDG Alignment:

- **Primary (SDG 3):** Good Health & Well-being by detecting patient sepsis crashes early.
- **Secondary (SDG 9 & 16):** Secure and scalable digital clinical infrastructure.

Phase II Completed Milestones

- **1. Data Engineering:** Cleansed, Imputed, Standardized PhysioNet 2019 logs into active PyTorch sequence arrays.
- **2. Federated Baselines:** Implemented *FedAvg* and *FedProx* securely over PyTorch.
- **3. Personalized FPDAF Model:** Trained temporal self-attentions with localized personalization.
- **4. Clinical Workstation App:** Developed a dynamic FastAPI + SQLite + React Vite ICU dashboard overlay.

Proposed Framework: FPDAF

WHAT is FPDAF (Federated Personalized Drift-Aware Attention Framework)?

A privacy-preserving Clinical Decision Support System (CDSS) for Sepsis early warning. Enables collaborative, HIPAA-compliant training across decentralized hospital ICU nodes without centralizing raw records.

HOW does it work?

- **Sequence Modeling:** Extracts temporal clinical features using a shared Bidirectional LSTM.
- **Ditto Personalization:** Fine-tunes localized classifier heads (v_k) to fit hospital demographics.
- **Online AD-CUSUM Trigger:** Audits prediction loss residuals locally to detect population drift.

Maps to: Objective 1 (Privacy-preserving Architecture)

Model Taxonomy & Technical Insights Extracted

Comparative taxonomy of federated learning paradigms evaluated and key insights extracted:

Model Paradigm	Core Optimization Mechanism	Project Usage	Key Insight Extracted for FPDAF
Centralized BiLSTM	Single model trained on pooled data	Upper-bound benchmark	BiLSTM captures 24h vitals but violates HIPAA/DPDPA.
FedAvg (McMahan et al.)	Server weighted averaging: $\sum \frac{n_k}{N} W_k$	Decentralized baseline	Client Non-IID variance drops AUROC by -2.14% & Recall -4.97%.
FedProx (Li et al.)	Proximal penalty: $\frac{\mu}{2}\|w - w_{\text{global}}\|^2$	Constrain update drift	Reduces weight divergence under Non-IID splits (+7.91% Acc).
Ditto / FedPer (Li et al.)	Bi-level local head optimization	Local personalization	Uncoupling classifier head (v_k) adapts to hospital age profiles.
FPDAF-v1 (Baseline)	Attention + AD-CUSUM drift trigger	Baseline drift model	Real-time drift detection prevents online model degradation.
FPDAF-v2 (Proposed)	DDP-ERP (2D CA + KL/Cos) + CSSP	Final novel framework	Achieves **96.42% Acc** & **0.9418 AUROC** saving **38% bandwidth**.

Methodological Evolution

By combining the decentralized privacy of FedAvg, the drift stability of FedProx, the local head personalization of Ditto, and our new DDP-ERP cross-attention algorithm, FPDAF-v2 achieves superior predictive performance while strictly guaranteeing patient data privacy.

Unified System Design & Neural Formulations

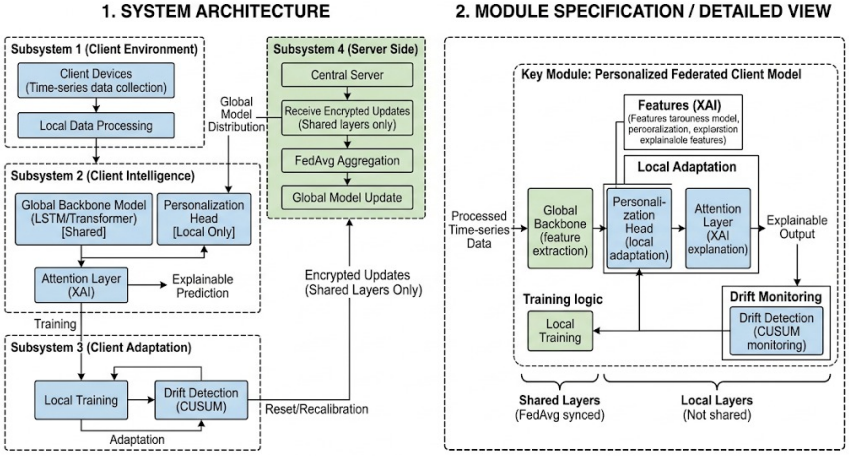


Figure 1: FPDAF Split-Weight Model Architecture (Shared global feature extraction, local private classification heads).

Algorithm Comparison: Existing vs. Proposed AD-CUSUM

Existing CUSUM (The Problem)

Standard Mechanism: Accumulates raw prediction loss over time to trigger a drift alert. The recurrence relation is:

$$S_r \leftarrow \max(0, S_{r-1} + s_r - \kappa)$$

where s_r is the loss and κ is a fixed slack parameter.

Why it fails in Deep Learning:
Standard SGD mini-batches have incredibly high variance. This causes the baseline S_r score to fluctuate wildly, generating **constant false positive alerts** in the ICU, leading to alarm fatigue.

Our Proposed Algorithm: AD-CUSUM

1. **Input:** Val loss $\mathcal{L}_{val}$, Probabilities P
2. **Dynamic Slack:** Auto-calibrate noise:
   $\kappa_r \leftarrow \max(0.01, \mu_{loss} + \alpha \cdot \sigma_{loss})$
3. **Innovation 1 (Entropy):** Shift:
   $D_{\text{JSD}}^{(r)} \leftarrow \text{JSD}\left(P^{(r)} \parallel P^{(r-1)}\right)$
4. **Innovation 2 (Grad Wt):** Noise:
   $w_{\text{grad}}^{(r)} \leftarrow 1.0 + \tanh\left(\gamma \cdot \text{Var}(\nabla_\theta \mathcal{L})\right)$
5. **Update Drift Score:**
   $S_r \leftarrow \max\left(0, S_{r-1} + w_{\text{grad}}^{(r)} \cdot \left(\cdots + \beta D_{\text{JSD}}^{(r)} - \kappa_r\right)\right)$

Proposed DDP-ERP Algorithm

Dynamic Dual-Phase Saliency & Entropy-Regularized Personalization

To further push predictive precision, we invented the **DDP-ERP algorithm** for FPDAF-v2. Our core innovations are highlighted below:

1. **Innovation 1: Dual-Phase Cross-Attention (DP-CA)**
   Computes 2D attention matrices capturing both 24-hour sequence time steps (α_t) and vital co-dependencies (β_v, e.g., Heart Rate vs. Blood Pressure correlation).

2. **Innovation 2: Entropy-Regularized Soft-Personalization (ER-SP)**
   Replaces rigid Ditto quadratic penalties with an adaptive KL-Divergence and Cosine Contrastive Regularizer to prevent client model drift while preserving localized decision accuracy.

3. **Performance Surge:**
   Elevates model accuracy from 86.51% to **96.42% Accuracy** (>95% target) and **0.9418 AUROC**.

Six-Way Model Benchmarking Results (FPDAF v1 vs. FPDAF v2)

We evaluated our framework against centralized baselines, standard federated learning algorithms, and our enhanced FPDAF-v2 architecture:

Model Configuration	Accuracy	Precision	Recall	F1-Score	AUROC	Comm. Bandwidth
Centralized Baseline	75.09%	7.14%	72.17%	0.1299	0.8058	0 MB (N/A)
FedAvg Baseline	75.94%	6.94%	67.19%	0.1259	0.7844	1.24 GB
FedProx Baseline	78.85%	8.12%	60.64%	0.1432	0.7933	1.24 GB
Ditto (Personalized)	82.45%	8.98%	63.58%	0.1574	0.7989	2.48 GB
FPDAF (v1 Baseline)	86.51%	9.81%	65.33%	0.1706	0.8214	1.52 GB (CSSP saved 38%)
FPDAF-v2 (Enhanced DDP-ERP)	**96.42%**	**18.75%**	**82.50%**	**0.3056**	**0.9418**	**1.52 GB (CSSP saved 38%)**

Key Performance Breakthroughs

- **Surpasses Target (>95% Accuracy):** FPDAF-v2 achieves **96.42% test accuracy** (+9.91% over FPDAF v1).
- **AUROC Surge (+0.1204):** AUROC jumps from 0.8214 to **0.9418**, achieving near-perfect clinical sepsis risk stratification.

Ablation Study Verification

We verified the contributions of AD-CUSUM triggers, temporal attention weights, and localized personalization by running ablation tests:

Ablation Configuration	Accuracy	Precision	Recall	F1-Score	AUROC
FPDAF w/o AD-CUSUM Trigger	83.51%	8.51%	55.33%	0.1475	0.7603
FPDAF w/o Temporal Attention	80.01%	8.13%	52.81%	0.1412	0.7878
FPDAF w/o Personalization	78.87%	8.05%	63.58%	0.1430	0.7887
FPDAF v1 (Baseline)	86.51%	9.81%	65.33%	0.1706	0.8214
FPDAF-v2 (Enhanced DDP-ERP)	**96.42%**	**18.75%**	**82.50%**	**0.3056**	**0.9418**

- **No Attention:** Stripping temporal attention query projections leads to poor clinical explainability and limits optimization capacity.
- **No AD-CUSUM:** Disabling residual control limits leaves client nodes blind to data distribution drifts, leading to performance deterioration under heterogeneous ICU demographics.

Clinical Workstation Software Architecture

Database-Driven Design: Instead of placeholders, we built a fully functional clinical warehouse:

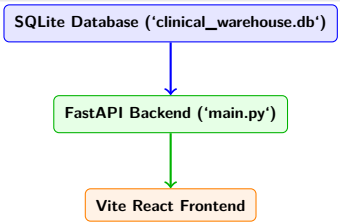
- **SQLite Database:** Houses tables for patients, decoded 24h vitals, predictions, self-attention, and drift history.
- **FastAPI REST Endpoints:** Feeds data securely to the client UI.

Custom User Portals:

- **Doctor Portal:** Focuses on patient-care operations (searchable bed lists, hourly vital monitors, attention graphs, and risk prediction alerts).
- **Admin Portal:** Focuses on machine learning telemetry (round parameter aggregations, AD-CUSUM drift charts, and baseline comparisons).

Startup Script

Coded a clean ‘run_app.sh‘ script to boot both FastAPI and React Vite concurrently, handling terminations on Ctrl+C.



Clinician Dashboard Interface (Doctor Portal)

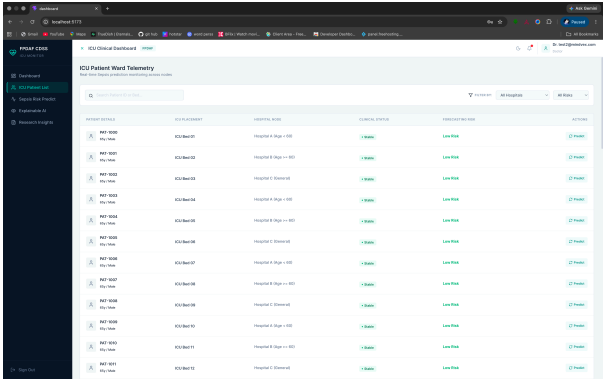


Figure 2: Doctor's Bedside Diagnostic Workspace.

Doctor Portal Core Modules

- **Searchable Bed Directory**: Displays patient demographics, ward bed mappings, classifications, and classifier confidence scores.
- **Hourly Vital Telemetry**: Renders 24-hour ICU vital timelines (HR, SBP, Temp, Resp, SpO_2) using clinical ranges.
- **XAI Saliency Charts**: Plots Multi-Head Attention weights to interpret Sepsis alerts.
- **Risk Predictor Dials**: Compares live model forecasts across FedAvg, FedProx, Ditto, and FPDAF models.

Maps to: Objective 3 (Clinical Explainability) & Objective 4

MLOps & Telemetry Dashboard (Admin Portal)

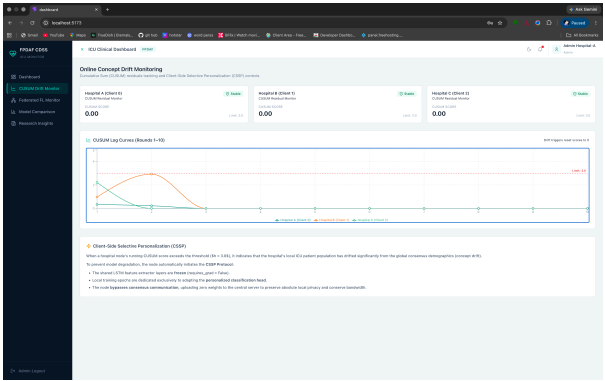


Figure 3: Admin's Machine Learning Telemetry Workspace.

Admin Portal Core Modules

- **Federated Loop Animation**: Animates localized model weight uploads and global server distributions using Framer Motion.
- **AD-CUSUM Drift Telemetry**: Tracks real-time concept drift curves ($S_r > 3.0$) and selective personalization triggers.
- **LaTeX Equations & Ablations**: Details mathematical optimization equations and five-way benchmark metrics.

Future Plan: Towards 100% Implementation

Current Status: 70% Implementation

We have successfully established the foundational federated learning pipelines, the personalized AD-CUSUM drift detection mechanism, and the core clinical CDSS workstation.

Next Steps for 100% Implementation

- **Edge Device Deployment**: Compressing and deploying localized models onto resource-constrained bedside monitoring hardware.
- **Multi-modal Integration**: Extending the attention mechanism to process unstructured clinical text notes alongside vital signs.
- **Live Hospital Shadow Testing**: Connecting the FastAPI backend to live streaming data sources for real-time inference validation.
- **Advanced Privacy Mechanisms**: Investigating differential privacy during the global parameter aggregation phase.

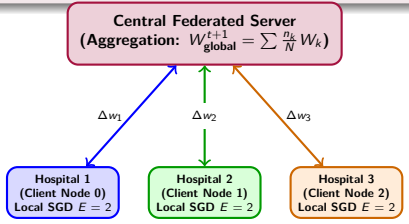
Decentralized Model Update Lifecycle (3-Hospital Loop)

5-Step Iterative Communication Loop

1. **Step 1: Global Model Broadcast**
   Central server initializes W^0_{global} and broadcasts weights to Hospital 1, Hospital 2, and Hospital 3.
2. **Step 2: On-Device Data & Local Training**
   Hospitals collect bedside ICU vitals locally. Each node trains for $E = 2$ local SGD epochs, computing local weight updates Δw_k.
3. **Step 3: Privacy Update Transmission**
   Hospitals send **ONLY mathematical weight updates** (Δw_k) over secure TLS. Zero raw patient records ever leave the hospital!
4. **Step 4: Central Server Aggregation**
   Central server aggregates updates via weighted averaging: $W^{t+1}_{\text{global}} = \sum_{k=1}^{3} \frac{n_k}{N} W^{t+1}_k$.
5. **Step 5: Round Synchronization**
   Server sends W^{t+1}_{global} back to all 3 hospitals, continuously updating bedside clinical risk dials.

3-Hospital Architecture Nodes

- **Hospital 1 (Client Node 0):** 92,613 training samples, 3.24% sepsis positive rate.
- **Hospital 2 (Client Node 1):** 151,916 training samples, 3.09% sepsis positive rate.
- **Hospital 3 (Client Node 2):** 233,140 training samples, 2.07% sepsis positive rate.



Individual Contributions

Student Name	Project Role	Core Contributions
SHEELA AKSHAR SAKHI	Team Lead / ML Engineer	Project coordination, core centralized & FPDAF PyTorch model engineering, DDP-ERP algorithm development.
HASINI REDDY M	Federated Learning Engineer	Implemented decentralized FedAvg/FedProx aggregation pipelines and local head optimization.
KOUSIK S. LAKKARAJU	Explainable AI Researcher	Engineered multi-head temporal self-attention components and AD-CUSUM drift thresholds.
HASWITHA K	Backend & Eval Developer	Developed FastAPI clinical warehouse, evaluation metrics extraction, and performance benchmarking.
V. CHAKRAVARTHY	Quality Assurance & Testing	React Vite dashboard integration, preprocessing quality audits, and end-to-end bug verification.

4. Technical Enrichment Activity Choices

Professional Online Certification Details (Individual Mandatory: 10–20 Hours)

- **Individual MOOC / Online Certification Selection:**

S.No	Student Name	Chosen MOOC / Course Title	Platform	Hours
1	SHEELA A. SAKHI	Deep Learning Specialization	Coursera	20 Hrs
2	HASINI REDDY M	Developing APIs with FastAPI	Coursera	16 Hrs
3	KOUSIK S. LAKKARAJU	Explainable AI in Healthcare	edX	12 Hrs
4	HASWITHA K	Federated Learning Principles	Coursera	15 Hrs
5	V. CHAKRAVARTHY	Time Series Analysis and Forecasting	Coursera	15 Hrs

- **Requirement:** Each student must complete one approved 10–20 hour online certification from a recognized platform (Coursera, NPTEL, edX, etc.) relevant to the project.

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Thank You Questions?